Drawing Analysis Results

AI Analysis Summary

Plan View: R2.0 (page 2)

Scale: AS NOTED

roof_drains: 3 | mechanical_units: 2 | sleeper_curbs: 4 | vent_hoods: 2

Detail refs: Detail 1/R3.2, Detail 5/R3.0, Detail 2/R3.2, Detail 1/R3.2, Detail 5/R3.1, Detail 4/R3.1, Detail 4/R3.0, Detail 6/R3.1, Detail 1/R3.1, Detail 3/R3.1, Detail 6/R3.0, Detail 4/R3.1, Detail 5/R3.1

Details: R2.1 (page 3)

> **SECTION #3 - ROOF SLOPE PLAN** [slope_plan] - measured in: sqft
>
> - 1. 2% SLOPE PACKAGE / TAPER PACKAGE → Polyisocyanurate_ISO_Insulation (2.4m (approx 94.5 inches) wide taper package at 2% slope around full perimeter of roof)

Details: R3.0 (page 4)

> **TYPICAL PROTECTED ROOF ASSEMBLY - NO TAPERED PACKAGE** [field_assembly] - measured in: sqft
>
> - 1. STRUCTURAL CONCRETE SLAB → None
> - 2. EXISTING ROOF MEMBRANE → Base_Membrane (existing)
> - 3. NEW ROOF BASE SHEET → Base_Membrane
> - 4. ROOF CAP SHEET → Cap_Membrane
> - 5. 10mm DRAINAGE MAT → Drainage_Board (10mm)
> - 6. 75mm EXTRUDED POLYSTYRENE INSULATION (XPS) → XPS_Insulation (75mm)
> - 7. FILTER FABRIC → Filter_Fabric
> - 8. REMOVE AND REPLACE GRAVEL BALLAST TO MAXIMUM 100mm → Gravel_Ballast (remove and replace, maximum 100mm)

> **TYPICAL PROTECTED ROOF ASSEMBLY - WITH TAPERED INSULATION** [field_assembly] - measured in: sqft
>
> - 1. STRUCTURAL CONCRETE DECK → None
> - 2. EXISTING MEMBRANE → Base_Membrane (existing)
> - 3. TAPERED INSULATION PACKAGE AS SHOWN ON DRAWINGS. ADHERED WITH ADHESIVE RIBBONS 300mm (12") O/C → Polyisocyanurate_ISO_Insulation (Adhered with adhesive ribbons 300mm o/c)
> - 4. COMBINATION BOARD BASE SHEET AND 6.4mm ASPHALT CORE BOARD ADHERED WITH ADHESIVE RIBBONS 300mm (12") O/C → Base_Membrane (6.4mm asphalt core board)
> - 5. CAP SHEET MEMBRANE → Cap_Membrane

- 6. 10mm DRAINAGE MAT → Drainage_Board (10mm)
- 7. 75mm EXTRUDED POLYSTYRENE INSULATION (XPS) → XPS_Insulation (75mm)
- 8. FILTER FABRIC → Filter_Fabric
- 9. REMOVE AND REPLACE GRAVEL BALLAST TO MAXIMUM 100mm → Gravel_Ballast (remove and replace, maximum 100mm)

TYPICAL ROOF DRAIN INSERT WATERPROOFING DETAIL [drain] - measured in: each

- 1. EXISTING STRUCTURAL CONCRETE → None (existing)
- 2. EXISTING ROOF DRAIN TO REMAIN → Roof_Drain (existing to remain)
- 3. SET DRAIN INTO PERIMETER SEALANT → Sealant_General
- 4. FILL VOID WITH MINERAL WOOL INSULATION → Batt_Insulation
- 5. INSTALL NEW DRAIN INSERT INTO EXISTING PIPE. SUPPLY INSERT WITH 150mm FLANGE → Roof_Drain (150mm flange)
- 6. DRAIN DOME → Roof_Drain
- 7. SECURE MEMBRANE WITH CLAMPING RING SYM. → Clips
- 8. APPLY FLEECE REINFORCED ALSAN RS MEMBRANE AROUND ROOF DRAIN. MINIMUM DIMENSION OF 1500x1500mm → Fleece_Reinforcement_Fabric (1500x1500mm minimum dimension)
- 9. PROVIDE NEW GRAVEL GUARD AROUND DRAIN → Flashing_General

ROOF AREA DIVIDER CURB WITH EXPANSION JOINT [expansion_joint] - measured in: linear_ft

- 1. EXISTING ROOF ASSEMBLY → None (existing)
- 2. PROVIDE NEW DIMENSIONAL LUMBER TO ACCOMMODATE NEW INSULATION → Wood_Blocking_Lumber
- 3. PROVIDE NEW 25MM WIDE POLYISOCYANURATE INSULATION → Polyisocyanurate_ISO_Insulation (25mm wide)
- 4. NEW 13mm PLYWOOD SHEATHING. SCREW AT TOP AND BOTTOM AT 300mm O/C → Plywood_Sheathing (13mm)
- 5. LEAVE A LOOP IN BASE AND CAP SHEET AT EXPANSION JOINT → Base_Membrane
- 6. INFILL JOINT WITH MINERAL WOOL → Batt_Insulation
- 7. REMOVE AND REINSTATE METAL CAP FLASHING → Coated_Metal_Sheet (remove and reinstate)
- 8. NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING. FINISH TIGHT TO INSULATION BELOW BALLAST → Coated_Metal_Sheet (24GA)

TYPICAL CONCRETE PANEL DETAIL [parapet] - measured in: linear_ft

- 1. CONCRETE DECK TO WALL → None
- 2. INSTALL FOAM BACKER ROD BELOW AIR/VAPOUR BARRIER → Sealant_General
- 3. NEW VAPOUR BARRIER AT TRANSITION FROM CONCRETE DECK TO WALL → Vapour_Barrier_Membrane
- 4. 26GA GALVANIZED STEEL METAL IN L SHAPE. FASTEN INTO PRE-CAST AT 300mm WITH 6.4mm DIAMETER PIN BOLTS → Coated_Metal_Sheet (26GA, 6.4mm pin bolts)
- 5. BASE AND CAP SHEET MEMBRANE TRANSITION FROM TOP OF CURB TO WALL. LEAVE LOOSE LOOP, LAY MEMBRANE HORIZONTALLY TO MINIMIZE JOINTS → Base_Membrane
- 6. TERMINATE MEMBRANE AT UPTURN WITH NEW 25mm CONT. CLAMPING BAR. FASTEN WITH 38mm HEX HEAD SCREWS AT 400mm O/C. → Flashing_General (25mm clamping bar)
- 7. PRECAST PANEL TO REMAIN IN PLACE → None (existing to remain)
- 8. PROVIDE NEW 18 ga STEEL STUD FRAMING C/W TOP AND BOTTOM TRACKS. ANCHOR TO DECK WITH SMACK PINS, 38mm LENGTH WITH MINIMUM 6mm EMBEDMENT INTO CONCRETE, INSTALL AT 300mm O/C IN STAGGERED W PATTERN FASTEN STUDS TO TRACKS WITH #10 SELF-TAPPING PAN HEAD SCREWS → Fasteners (18 ga steel stud framing)

- 9. 12mm GYPSUM SHEATHING MECHANICALLY FASTENED INTO STRUCTURE OF CURB. USE 50mm SELF TAPPING CORROSION RESISTANT #8 FLAT HEAD OR PAN HEAD SCREWS WITH 50mm PLATES C/W NEOPRENE GASKET → Gypsum_Fiber_Coverboard (12mm)
- 10. PROVIDE 24GA PRE-FINISHED STEEL FLASHING TO COVER TERMINATION BAR. FASTEN AT BASE WITH 6.4mm PIN BOLTS EMBEDDED 38mm SPACED AT MAX. 400mm O/C. FIT TOP INTO EXISTING REGLET AND FINISH WITH SEALANT → Coated_Metal_Sheet (24GA)

TYPICAL GAS PENETRATION [penetration_gas] - measured in: each

- 1. CONTINUE MEMBRANE BELOW NEW CURB → Base_Membrane
- 2. TWO PIECE FLAT STOCK 26GA STEEL CUT ON SITE TIGHT TO PIPE PENETRATION. FASTEN TO DECK WITH EIGHT 25mm ZAMAC PIN BOLTS EQUALLY SPACED. LAP MEMBRANE ONTO METAL MINIMUM 150mm(6"). → Coated_Metal_Sheet (26GA, 150mm lap)
- 3. PROVIDE FIRESTOP SEALANT BETWEEN PIPE AND SHEET METAL → Sealant_General
- 4. PROVIDE NEW DIMENSIONAL LUMBER AND PLYWOOD CURB AROUND PENETRATION. FILL ENTIRE CAVITY WITH MINERAL WOOL INSULATION. COVER STUDS WITH 13mm PLYWOOD SHEATHING AROUND PERIMETER AND TOP → Wood_Blocking_Lumber (13mm plywood sheathing)
- 5. INSULATE WITH MINERAL WOOL INSULATION → Batt_Insulation
- 6. PROVIDE SELF-ADHERED BASE SHEET AROUND CURB. DO NOT TORCH TO WOOD SHEATHING. → Base_Membrane_Peel_Stick
- 7. PROVIDE FLEECE REINFORCED LIQUID APPLIED FLASHING AROUND PENETRATION → Fleece_Reinforcement_Fabric
- 8. NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING. FINISH TIGHT TO INSULATION BELOW BALLAST → Coated_Metal_Sheet (24GA)
- 9. PROVIDE 22GA PREFINISHED STEEL COVER FLASHING OVER FULL CURB. ONE PIECE METAL SLEEVE. NO SHARP EDGES. COVER MUST BE REMOVABLE. FASTEN WITH FOUR HEX HEAD #8 WOOD SCREWS → Coated_Metal_Sheet (22GA)
- 10. REPAINT GAS LINES TO MATCH EXISTING → Coating_Paint
- 11. PROVIDE NEW SHUT-OFF VALVE IN ADDED PIPE SECTION → None
- 12. PROVIDE NEW RUBBER BLOCKING BELOW GAS LINE TO MATCH EXISTING. → Walkway_Pads

Details: R3.1 (page 5)

SMALL OPENING COVER [opening_cover] - measured in: each

- 1. 3mm STEEL PLATE COVER → Coated_Metal_Sheet (3mm thick)
- 2. TAPCON PHILLIPS HEAD, 44mm (1-3/4") LENGTH, 6mm (1/4"), 4 PER COUNTERSUNK PLATE → Fasteners (44mm (1-3/4") length, 6mm (1/4") diameter)

LARGE OPENING COVER [opening_cover] - measured in: each

- 1. 38x38x6.4 ANGLE FULLY AROUND PERIMETER → Flashing_General (38x38x6.4mm angle)
- 2. 38x38x6.4 STEEL ANGLE ON EDGE. SPACE AT 450 O/C MAX. IN SINGLE DIRECTION → Flashing_General (38x38x6.4mm angle)
- 3. 3mm STEEL PLATE COVER → Coated_Metal_Sheet (3mm thick)
- 4. TAPCON PHILLIPS HEAD 38mm, MIN. 8 PER OPENING → Fasteners (38mm length)

MECHANICAL SLEEPER CURB DETAIL [sleeper_curb] - measured in: linear_ft

- 1. FASTEN TO EXISTING SLAB WITH #8 CONCRETE SCREWS AT 150mm O/C → Fasteners (#8 concrete screws)
- 2. INSTALL NEW WOOD BLOCKING AS REQUIRED → Wood_Blocking_Lumber (Wood blocking)
- 3. NEW 13mm PLYWOOD SHEATHING → Plywood_Sheathing (13mm thick)
- 4. NEW 24GA PREFINISHED STEEL COUNTER FLASHING → Flashing_General (24ga steel)
- 5. NEW 22GA. PRE-FINISHED STEEL CAP FLASHING → Flashing_General (22ga steel)
- 6. FOUR 50mm HEX HEAD #14 FASTENERS WITH RUBBERIZED GASKET → Fasteners (50mm length, #14 fasteners)
- 7. SET FOOTING INTO SEALANT AND PROVIDE PERIMETER HEEL BEAD → Sealant_General (Sealant bead)
- 8. INSTALL 24ga GALVANIZED STEEL STRAPPING FROM LEG ONTO SIDE OF CURB ON EACH CORNER → Flashing_General (24ga steel strapping, 4 straps total)

TYPICAL MECHANICAL CURB [mechanical_curb] - measured in: linear_ft

- 1. NEW DIMENSIONAL LUMBER AS REQUIRED TO PROVIDE 200mm UPTURN → Wood_Blocking_Lumber (200mm upturn)
- 2. 6mm ASPHALTIC CORE BOARD OR 12mm GYPSUM SHEATHING → Gypsum_Fiber_Coverboard (12mm gypsum or 6mm core board)
- 3. 50mm SELF TAPPING #8 FLAT HEAD OR PAN HEAD SCREWS WITH 50mm PLATES → Fasteners (50mm length)
- 4. RETURN BASE MEMBRANE OVER CURB → Base_Membrane (Base membrane)
- 5. FASTEN WITH RING NAILS → Nails_Staples (Ring nails)
- 6. NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING → Flashing_General (24ga steel)
- 7. REMOVE AND REINSTATE MECHANICAL EQUIPMENT → HVAC_Curb_Detail (remove and reinstate mechanical equipment)

MECHANICAL UNIT TRANSITION DETAIL - TYPE 2 [mechanical_curb] - measured in: linear_ft

- 1. EXISTING BASE AND CAP SHEET TO REMAIN IN PLACE → Base_Membrane (existing base and cap sheet to remain in place)
- 2. NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING → Flashing_General (24ga steel)
- 3. FASTEN @ 300mm O/C WITH HEX HEAD 50mm CONCRETE SCREWS C/W GASKET → Fasteners (50mm length, 300mm O/C)
- 4. EXISTING FLASHING TO REMAIN → Flashing_General (existing flashing to remain)
- 5. APPLY FLEECE REINFORCED PMMA MEMBRANE AT EXISTING SEAMS AND FROM FLASHING TO STEEL CHANNEL → Fleece_Reinforcement_Fabric (Fleece reinforced PMMA)
- 6. EXISTING MECHANICAL UNIT TO REMAIN → HVAC_Curb_Detail (existing mechanical unit to remain)

MECHANICAL UNIT TRANSITION DETAIL - TYPE 1 [mechanical_curb] - measured in: linear_ft

- 1. NEW BASE AND CAP SHEET INSTALLED UP UNDER THE EXISTING MECHANICAL UNIT FRAME → Base_Membrane (Base and cap sheet)
- 2. NEW 24GA PRE-FINISHED STEEL FLASHING → Flashing_General (24ga steel)
- 3. FASTEN @ 300mm O/C WITH HEX HEAD 50mm CONCRETE SCREWS C/W GASKET → Fasteners (50mm length, 300mm O/C)
- 4. APPLY SEALANT BEAD AT LEADING EDGE → Sealant_General (Sealant bead)
- 5. APPLY FLEECE REINFORCED PMMA MEMBRANE AT EXISTING SEAMS AND FROM FLASHING TO STEEL CHANNEL → Fleece_Reinforcement_Fabric (Fleece reinforced PMMA)

Details: R3.2 (page 6)

> **ROOF TO CURTAIN WALL DETAIL** [curtain_wall] - measured in: linear_ft
>
> - 1. NEW 12.5mm PLYWOOD ON EXTERIOR → Plywood_Sheathing (12.5mm thickness)
> - 2. NEW 26GA METAL BREAKSHAPE TO TRANSITION FROM CURB TO CURTAIN WALL FRAMING → Flashing_General (26GA)
> - 3. MINERAL WOOL INSULATION → Batt_Insulation (Fill gaps)
> - 4. NEW SOPREMA 1100HT SELF-ADHERED MEMBRANE CURTAIN WALL SHOULDER → Base_Membrane_Peel_Stick
> - 5. BEAUTY CAP AND ANTI-ROTATION BLOCK → None (remove and reinstate beauty cap and anti-rotation block, replace gaskets)
> - 6. NEW DRIP FLASHING AT CURTAIN WALL → Drip_Edge (Capture by pressure plate and anti-rotation block)
> - 7. NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING → Flashing_General (24GA; Finish tight to insulation below ballast; 100mm (4") overlap, 50mm (2") height shown)

> **TYPICAL PARAPET DETAIL** [parapet] - measured in: linear_ft
>
> - 1. EXISTING PRE-CAST CONCRETE EXTERIOR → None (existing to remain)
> - 2. NEW DIMENSIONAL LUMBER AND WOOD BLOCKING TO BUILD UP PARAPET → Wood_Blocking_Lumber (Fasten into concrete parapet with hex head #8 concrete screws in 'W' pattern at 300mm O/C)
> - 3. NEW 12.5mm PLYWOOD SHEATHING ON INTERIOR FACE OF PARAPET WALL → Plywood_Sheathing (12.5mm thickness; Fasten with 50mm pin bolts at max 300mm O/C vertically and 600mm O/C horizontally)
> - 4. CONTINUOUS 22GA GALV. STEEL CLIP ON EXTERIOR FACE → Clips (22GA; Fasten at 400mm O/C with 50mm #8 pan head screws in a staggered pattern min. 50mm vertical spacing)
> - 5. 24GA PRE-FINISHED STEEL COUNTER FLASHING → Flashing_General (24GA; Max 2.4m lengths; Utilize S-locks on vertical joints)
> - 6. 24GA PREFINISHED STEEL CAP FLASHING → Flashing_General (24GA; Face fasten interior at 400mm O/C with 50mm #12 hex head concrete screws; Slope 5% min.)
> - 7. STANDING SEAM AT CORNERS AND S-LOCK IN CAP FLASHING → Standing_Seam_Metal

Specification PDF Analysis (Deterministic Regex Engine)

24 unique products found across **7** categories — **11** mapped to pricing keys.
Project: 2026 Roof Replacement

▶ **Membranes (8)**

▶ **Sealants & Adhesives (4)**

▶ **Drainage (4)**

▶ **Miscellaneous (3)**

▶ **Fasteners & Hardware (1)**

▶ **Insulation (1)**

▶ Metal Flashings & Accessories (3)

Spec-Driven Pricing Estimate (Drawing spatial data + Specification materials — deterministic)

LINE ITEMS RESOLVED	RESOLUTION FAILURES	TOTAL MATERIAL COST
8	6	$70,026.60

Resolved Line Items

DETAIL	TYPE	CONFIRMED MATERIAL	QTY	UNIT PRICE	LINE COST	SOURCE
TYPICAL PROTECTED ROOF ASSEMBLY - NO TAPERED PACKAGE R3.0	field_assembly	Drainage Board `Drainage_Board` spec pg 53	18 roll (6'x50')	$410.53	$7,389.54	FAL
TYPICAL PROTECTED ROOF ASSEMBLY - NO TAPERED PACKAGE R3.0	field_assembly	Filter Fabric `Filter_Fabric` spec pg 53, 58	5269 sqft	$0.25	$1,317.25	FAL
TYPICAL PROTECTED ROOF ASSEMBLY - NO TAPERED PACKAGE R3.0	field_assembly	Gravel Ballast `Gravel_Ballast` spec pg 48	5269 sqft	$0.75	$3,951.75	FAL
TYPICAL PROTECTED ROOF ASSEMBLY - NO TAPERED PACKAGE R3.0	field_assembly	SBS Membrane `SBS_Membrane` spec pg 49	53 roll	$298.00	$15,794.00	FAL
				Total Material Cost	**$70,026.60**	

DETAIL	TYPE	CONFIRMED MATERIAL	QTY	UNIT PRICE	LINE COST	SOURCE
TYPICAL PROTECTED ROOF ASSEMBLY - NO TAPERED PACKAGE R3.0	field_assembly	2-Ply SBS Membrane System `SBS_2Ply_Modified_Bitumen` spec pg 30	5269 sqft	$7.00	$36,883.00	FAL
TYPICAL ROOF DRAIN INSERT WATERPROOFING DETAIL R3.0	drain	Sealant (General) `Sealant_General` spec pg 2, 11, 14, 30, 48, 57, 64, 68, 69, 70, 71, 72	3 tube	$29.86	$89.58	FAL
TYPICAL ROOF DRAIN INSERT WATERPROOFING DETAIL R3.0	drain	Roof Drain `Roof_Drain` spec pg 2, 49, 59, 73, 74	3 EA	$325.00	$975.00	FAL
TYPICAL PARAPET DETAIL R3.2	parapet	Standing Seam Metal `Standing_Seam_Metal` spec pg 63, 66	572 lin ft	$6.34	$3,626.48	FAL
				Total Material Cost	**$70,026.60**	

⚠ Material Resolution Failures (6)

These drawing details could not be matched to a confirmed material in the Specification PDF. Either the spec PDF does not cover these items, or additional regex patterns are needed in `database.PRODUCT_KEYWORDS`.

- **ROOF AREA DIVIDER CURB WITH EXPANSION JOINT** expansion_joint (ref: R3.0)

  Expected one of: `Wood_Blocking_Lumber`, `Polyisocyanurate_ISO_Insulation`, `Plywood_Sheathing`, `Base_Membrane`, `Batt_Insulation`, `Coated_Metal_Sheet`, `Flashing_General`, `EPDM_Accessory`, `TPO_Accessory` — none confirmed in spec.

- **SMALL OPENING COVER** opening_cover (ref: R3.1)

  Expected one of: `Coated_Metal_Sheet`, `Fasteners`, `Plywood_Sheathing`, `Flashing_General` — none confirmed in spec.

- **LARGE OPENING COVER** opening_cover (ref: R3.1)

  Expected one of: `Flashing_General`, `Coated_Metal_Sheet`, `Fasteners`, `Plywood_Sheathing` — none confirmed in spec.

- **TYPICAL MECHANICAL CURB** mechanical_curb (ref: R3.1)

  Expected one of: `Wood_Blocking_Lumber`, `Gypsum_Fiber_Coverboard`, `Fasteners`, `Base_Membrane`, `Nails_Staples`, `Flashing_General`,

`Metal_Flashing_Galvanized` — none confirmed in spec.

- **MECHANICAL UNIT TRANSITION DETAIL - TYPE 2** mechanical_curb (ref: R3.1)

  Expected one of: `Base_Membrane`, `Flashing_General`, `Fasteners`, `Fleece_Reinforcement_Fabric`, `HVAC_Curb_Detail`, `Metal_Flashing_Galvanized` — none confirmed in spec.

- **ROOF TO CURTAIN WALL DETAIL** curtain_wall (ref: R3.2)

  Expected one of: `Plywood_Sheathing`, `Flashing_General`, `Batt_Insulation`, `Base_Membrane_Peel_Stick`, `Drip_Edge`, `Metal_Flashing_Galvanized`, `Metal_Flashing_Prepainted` — none confirmed in spec.

Detail-Based Quantity Takeoff (AI-Analyzed)

ROOF AREA	PERIMETER	PARAPET	PENETRATIONS
9,580 sqft	**519 LF**	**519 LF x 0.0 ft**	**8**

Field Assembly (Plan View) [field_assembly] (ref: ?)

Measured in: sqft | Base value: 9,580.0 | Detail cost: $62,176.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
Vapour Barrier (Elastophene SP 2.2)	66 roll (1m x 15m)	$82.45	$5,441.70
ISO Insulation 2.5" (Sopra-ISO)	659 sheet (4'x4')	$25.60	$16,870.40
Tapered ISO Insulation (drainage slope)	0 sqft	$3.10	-
Densdeck Coverboard 1/2"	330 sheet (4'x8')	$34.20	$11,286.00
SBS Base Sheet - Field (Sopraply Base 520)	111 roll	$127.25	$14,124.75
SBS Base Sheet - Wall (Sopraply Base 520)	3 roll	$127.25	$381.75
SBS Cap Sheet - Field (Sopraply Traffic Cap)	129 roll	$105.80	$13,648.20
SBS Cap Sheet - Wall (Sopraply Traffic Cap)	4 roll	$105.80	$423.20

TYPICAL PROTECTED ROOF ASSEMBLY - NO TAPERED PACKAGE [field_assembly] (ref: R3.0)

Alternative — not priced

Measured in: sqft | Base value: 9,580.0 | Detail cost: $0.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST

TYPICAL PROTECTED ROOF ASSEMBLY - WITH TAPERED INSULATION [field_assembly] (ref: R3.0) Alternative — not priced

Measured in: sqft | Base value: 9,580.0 | Detail cost: $0.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST

TYPICAL ROOF DRAIN INSERT WATERPROOFING DETAIL [drain] (ref: R3.0)

Measured in: each | Base value: 3 | Detail cost: $7,012.86 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
EXISTING STRUCTURAL CONCRETE	10538 unit	$0.00	-
SET DRAIN INTO PERIMETER SEALANT	29 tube	$29.86	$865.94
FILL VOID WITH MINERAL WOOL INSULATION	8 bundle	$83.99	$671.92
INSTALL NEW DRAIN INSERT INTO EXISTING PIPE. SUPPLY INSERT WITH 150mm FLANGE	3 EA	$325.00	$975.00
DRAIN DOME	0 EA	$0.00	-
SECURE MEMBRANE WITH CLAMPING RING SYM.	0 piece	$10.19	-
APPLY FLEECE REINFORCED ALSAN RS MEMBRANE AROUND ROOF DRAIN. MINIMUM DIMENSION OF 1500x1500mm	1 roll	$150.00	$150.00
PROVIDE NEW GRAVEL GUARD AROUND DRAIN	58 10ft piece	$75.00	$4,350.00

ROOF AREA DIVIDER CURB WITH EXPANSION JOINT [expansion_joint] (ref: R3.0)

Measured in: linear_ft | Base value: 129.8 | Detail cost: $35,043.60 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
EXISTING ROOF ASSEMBLY	0 unit	$0.00	-
PROVIDE NEW DIMENSIONAL LUMBER TO ACCOMMODATE NEW INSULATION	72 8ft piece	$16.59	$1,194.48
PROVIDE NEW 25MM WIDE POLYISOCYANURATE INSULATION	659 sheet (4'x4')	$43.62	$28,745.58
NEW 13mm PLYWOOD SHEATHING. SCREW AT TOP AND BOTTOM AT 300mm O/C	9 4'x8' sheet	$48.74	$438.66
LEAVE A LOOP IN BASE AND CAP SHEET AT EXPANSION JOINT	0 roll	$0.00	-
INFILL JOINT WITH MINERAL WOOL	0 bundle	$0.00	-
NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING. FINISH TIGHT TO INSULATION BELOW BALLAST	8 sheet (4'x10')	$583.11	$4,664.88

TYPICAL CONCRETE PANEL DETAIL [parapet] (ref: R3.0)

Measured in: linear_ft | Base value: 519.3 | Detail cost: $1,663.90 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
CONCRETE DECK TO WALL	0 unit	$0.00	-
INSTALL FOAM BACKER ROD BELOW AIR/VAPOUR BARRIER	0 tube	$0.00	-
NEW VAPOUR BARRIER AT TRANSITION FROM CONCRETE DECK TO WALL	2 roll	$173.66	$347.32
26GA GALVANIZED STEEL METAL IN L SHAPE. FASTEN INTO PRE-CAST AT 300mm WITH 6.4mm DIAMETER PIN BOLTS	0 sheet (4'x10')	$0.00	-
BASE AND CAP SHEET MEMBRANE TRANSITION FROM TOP OF CURB TO WALL. LEAVE LOOSE LOOP, LAY MEMBRANE HORIZONTALLY TO MINIMIZE JOINTS	0 roll	$0.00	-
TERMINATE MEMBRANE AT UPTURN WITH NEW 25mm CONT. CLAMPING BAR. FASTEN WITH 38mm HEX HEAD SCREWS AT 400mm O/C.	0 10ft piece	$0.00	-
PROVIDE NEW 18 ga STEEL STUD FRAMING C/W TOP AND BOTTOM TRACKS. ANCHOR TO DECK WITH SMACK PINS, 38mm LENGTH WITH	3 box (1M)	$350.00	$1,050.00

MATERIAL	QTY	UNIT PRICE	LINE COST
MINIMUM 6mm EMBEDMENT INTO CONCRETE, INSTALL AT 300mm O/C IN STAGGERED W PATTERN FASTEN STUDS TO TRACKS WITH #10 SELF-TAPPING PAN HEAD SCREWS			
12mm GYPSUM SHEATHING MECHANICALLY FASTENED INTO STRUCTURE OF CURB. USE 50mm SELF TAPPING CORROSION RESISTANT #8 FLAT HEAD OR PAN HEAD SCREWS WITH 50mm PLATES C/W NEOPRENE GASKET	9 sheet (4'x8')	$29.62	$266.58
PROVIDE 24GA PRE-FINISHED STEEL FLASHING TO COVER TERMINATION BAR. FASTEN AT BASE WITH 6.4mm PIN BOLTS EMBEDDED 38mm SPACED AT MAX. 400mm O/C. FIT TOP INTO EXISTING REGLET AND FINISH WITH SEALANT	0 sheet (4'x10')	$0.00	-

TYPICAL GAS PENETRATION [penetration_gas] (ref: R3.0)

Measured in: each | Base value: 1.0 | Detail cost: $1,843.27 qty: unit_perimeter G

Unit Breakdown:

- G-1: 1.57 LF perimeter (near grid line 3)

MATERIAL	QTY	UNIT PRICE	LINE COST
CONTINUE MEMBRANE BELOW NEW CURB	0 roll	$0.00	-
TWO PIECE FLAT STOCK 26GA STEEL CUT ON SITE TIGHT TO PIPE PENETRATION. FASTEN TO DECK WITH EIGHT 25mm ZAMAC PIN BOLTS EQUALLY SPACED. LAP MEMBRANE ONTO METAL MINIMUM 150mm(6").	0 sheet (4'x10')	$0.00	-
PROVIDE FIRESTOP SEALANT BETWEEN PIPE AND SHEET METAL	0 tube	$0.00	-
PROVIDE NEW DIMENSIONAL LUMBER AND PLYWOOD CURB AROUND PENETRATION. FILL ENTIRE CAVITY WITH MINERAL WOOL INSULATION. COVER STUDS WITH 13mm PLYWOOD SHEATHING AROUND PERIMETER AND TOP	0 8ft piece	$0.00	-
INSULATE WITH MINERAL WOOL INSULATION	0 bundle	$0.00	-
PROVIDE SELF-ADHERED BASE SHEET AROUND CURB. DO NOT TORCH TO WOOD SHEATHING.	1 roll	$165.00	$165.00

MATERIAL	QTY	UNIT PRICE	LINE COST
PROVIDE FLEECE REINFORCED LIQUID APPLIED FLASHING AROUND PENETRATION	0 roll	$0.00	-
NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING. FINISH TIGHT TO INSULATION BELOW BALLAST	0 sheet (4'x10')	$0.00	-
PROVIDE 22GA PREFINISHED STEEL COVER FLASHING OVER FULL CURB. ONE PIECE METAL SLEEVE. NO SHARP EDGES. COVER MUST BE REMOVABLE. FASTEN WITH FOUR HEX HEAD #8 WOOD SCREWS	0 sheet (4'x10')	$0.00	-
REPAINT GAS LINES TO MATCH EXISTING	1 pail	$1,088.17	$1,088.17
PROVIDE NEW SHUT-OFF VALVE IN ADDED PIPE SECTION	0 unit	$0.00	-
PROVIDE NEW RUBBER BLOCKING BELOW GAS LINE TO MATCH EXISTING.	1 pad	$590.10	$590.10

SMALL OPENING COVER [opening_cover] (ref: R3.1)

Measured in: each | Base value: 2 | Detail cost: $0.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
3mm STEEL PLATE COVER	0 sheet (4'x10')	$0.00	-
TAPCON PHILLIPS HEAD, 44mm (1-3/4") LENGTH, 6mm (1/4"), 4 PER COUNTERSUNK PLATE	0 box (1M)	$0.00	-

LARGE OPENING COVER [opening_cover] (ref: R3.1)

Measured in: each | Base value: 2 | Detail cost: $0.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
38x38x6.4 ANGLE FULLY AROUND PERIMETER	0 10ft piece	$0.00	-

MATERIAL	QTY	UNIT PRICE	LINE COST
38x38x6.4 STEEL ANGLE ON EDGE. SPACE AT 450 O/C MAX. IN SINGLE DIRECTION	0 10ft piece	$0.00	-
3mm STEEL PLATE COVER	0 sheet (4'x10')	$0.00	-
TAPCON PHILLIPS HEAD 38mm, MIN. 8 PER OPENING	0 box (1M)	$0.00	-

MECHANICAL SLEEPER CURB DETAIL [sleeper_curb] (ref: R3.1)

Measured in: each | Base value: 4 | Detail cost: $0.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
FASTEN TO EXISTING SLAB WITH #8 CONCRETE SCREWS AT 150mm O/C	0 box (1M)	$0.00	-
INSTALL NEW WOOD BLOCKING AS REQUIRED	0 8ft piece	$0.00	-
NEW 13mm PLYWOOD SHEATHING	0 4'x8' sheet	$0.00	-
NEW 24GA PREFINISHED STEEL COUNTER FLASHING	0 10ft piece	$0.00	-
NEW 22GA. PRE-FINISHED STEEL CAP FLASHING	0 10ft piece	$0.00	-
FOUR 50mm HEX HEAD #14 FASTENERS WITH RUBBERIZED GASKET	0 box (1M)	$0.00	-
SET FOOTING INTO SEALANT AND PROVIDE PERIMETER HEEL BEAD	0 tube	$0.00	-
INSTALL 24ga GALVANIZED STEEL STRAPPING FROM LEG ONTO SIDE OF CURB ON EACH CORNER	0 10ft piece	$0.00	-

TYPICAL MECHANICAL CURB [mechanical_curb] (ref: R3.1)

Measured in: linear_ft | Base value: 16.0 | Detail cost: $29.75 qty: unit_perimeter V

Unit Breakdown:

- V-1: 8.0 LF perimeter (near grid line 3)
- V-2: 8.0 LF perimeter (near grid line 4)

MATERIAL	QTY	UNIT PRICE	LINE COST
NEW DIMENSIONAL LUMBER AS REQUIRED TO PROVIDE 200mm UPTURN	0 8ft piece	$0.00	-
6mm ASPHALTIC CORE BOARD OR 12mm GYPSUM SHEATHING	0 sheet (4'x8')	$0.00	-
50mm SELF TAPPING #8 FLAT HEAD OR PAN HEAD SCREWS WITH 50mm PLATES	0 box (1M)	$0.00	-
RETURN BASE MEMBRANE OVER CURB	0 roll	$0.00	-
FASTEN WITH RING NAILS	1 box	$29.75	$29.75
NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING	0 10ft piece	$0.00	-

MECHANICAL UNIT TRANSITION DETAIL - TYPE 2 [mechanical_curb] (ref: R3.1)

Measured in: linear_ft | Base value: 28.0 | Detail cost: $1,700.00 qty: unit_perimeter RTU #2

Unit Breakdown:

- RTU #2-1: 28.0 LF perimeter (near grid line 3)

MATERIAL	QTY	UNIT PRICE	LINE COST
EXISTING BASE AND CAP SHEET TO REMAIN IN PLACE	0 roll	$0.00	-
NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING	0 10ft piece	$0.00	-
FASTEN @ 300mm O/C WITH HEX HEAD 50mm CONCRETE SCREWS C/W GASKET	0 box (1M)	$0.00	-
EXISTING FLASHING TO REMAIN	0 10ft piece	$0.00	-

MATERIAL	QTY	UNIT PRICE	LINE COST
APPLY FLEECE REINFORCED PMMA MEMBRANE AT EXISTING SEAMS AND FROM FLASHING TO STEEL CHANNEL	0 roll	$0.00	-
EXISTING MECHANICAL UNIT TO REMAIN	2 EA	$850.00	$1,700.00

MECHANICAL UNIT TRANSITION DETAIL - TYPE 1 [mechanical_curb] (ref: R3.1)

Measured in: linear_ft | Base value: 28.0 | Detail cost: $0.00 qty: unit_perimeter RTU #1

Unit Breakdown:

- RTU #1-1: 28.0 LF perimeter (near grid line 4)

MATERIAL	QTY	UNIT PRICE	LINE COST
NEW BASE AND CAP SHEET INSTALLED UP UNDER THE EXISTING MECHANICAL UNIT FRAME	0 roll	$0.00	-
NEW 24GA PRE-FINISHED STEEL FLASHING	0 10ft piece	$0.00	-
FASTEN @ 300mm O/C WITH HEX HEAD 50mm CONCRETE SCREWS C/W GASKET	0 box (1M)	$0.00	-
APPLY SEALANT BEAD AT LEADING EDGE	0 tube	$0.00	-
APPLY FLEECE REINFORCED PMMA MEMBRANE AT EXISTING SEAMS AND FROM FLASHING TO STEEL CHANNEL	0 roll	$0.00	-

ROOF TO CURTAIN WALL DETAIL [curtain_wall] (ref: R3.2)

Measured in: linear_ft | Base value: 519.3 | Detail cost: $397.30 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
NEW 12.5mm PLYWOOD ON EXTERIOR	0 4'x8' sheet	$0.00	-

MATERIAL	QTY	UNIT PRICE	LINE COST
NEW 26GA METAL BREAKSHAPE TO TRANSITION FROM CURB TO CURTAIN WALL FRAMING	0 10ft piece	$0.00	-
MINERAL WOOL INSULATION	0 bundle	$0.00	-
NEW SOPREMA 1100HT SELF-ADHERED MEMBRANE CURTAIN WALL SHOULDER	0 roll	$0.00	-
NEW DRIP FLASHING AT CURTAIN WALL	58 10ft piece	$6.85	$397.30
NEW 24GA PRE-FINISHED STEEL COUNTER FLASHING	0 10ft piece	$0.00	-

TYPICAL PARAPET DETAIL [parapet] (ref: R3.2)

Measured in: linear_ft | Base value: 519.3 | Detail cost: $3,626.48 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
NEW DIMENSIONAL LUMBER AND WOOD BLOCKING TO BUILD UP PARAPET	0 8ft piece	$0.00	-
NEW 12.5mm PLYWOOD SHEATHING ON INTERIOR FACE OF PARAPET WALL	0 4'x8' sheet	$0.00	-
CONTINUOUS 22GA GALV. STEEL CLIP ON EXTERIOR FACE	0 piece	$0.00	-
24GA PRE-FINISHED STEEL COUNTER FLASHING	0 10ft piece	$0.00	-
24GA PREFINISHED STEEL CAP FLASHING	0 10ft piece	$0.00	-
STANDING SEAM AT CORNERS AND S-LOCK IN CAP FLASHING	572 lin ft	$6.34	$3,626.48

Total Material Cost	**$113,493.16**

Standard Estimate (for comparison)

Standard Total Estimate	**$166,878.64**